



Architectural Decisions



Architectural Decision

A core feature of the NutriLog application is to allow the patient to search for and add foods to their meals (e.g., "White rice," "Fuji Apple," "Chicken Breast"). To make this work, the system requires access to a massive database containing thousands of food items and their respective nutritional information (calories, proteins, etc.). How we obtain, store, and access this data is a fundamental decision that will impact the system's performance, cost, and maintenance.



Architectural Decision

Should we create and maintain our own internal food database or integrate the system with an external API specialized in nutritional data?

Decision: Integrate with an external API specialized in nutritional data.

Architectural Decision



External Nutrition API (e.g., Open Food Facts, USDA FoodData Central, etc.):

What is it? Instead of storing the data locally, every time a patient searches for "Apple," our application sends a request to a third-party service via the internet. This service responds with a list of apples and their nutritional information.

Arguments in Favor:

- **Access to Rich and Maintained Data:** We gain instant access to a vast database that is curated and updated by subject matter experts.
- **Reduced Cost and Effort:** The responsibility for maintaining the data lies with the API provider, not our team. We focus on building our application, not on managing data.
- **Rapid Initial Development:** We can have the food search functionality up and running much faster.

Arguments Against:

- **Third-Party Dependency:** If the API goes down, our core functionality stops working. If the provider changes the terms, starts charging, or is discontinued, our system is directly impacted.
- **Latency:** Searches may be slower, as they depend on the user's internet speed and the API server's response time.
- **API Costs:** Many high-quality services charge based on request volume, which can become a significant operational cost as the user base grows.